Progressive Diffusion With Sparse Point Hypotheses for Moving Vehicle Detection From Satellite Videos

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***Abstract*—Detecting moving vehicles in satellite videos remains challenging due to the extremely small target size and the severe foreground–background imbalance. Most existing approaches extend convolution-based dense detectors to the temporal domain and rely on fixed grid-aligned prediction, which becomes inefficient and unstable under highly sparse foreground conditions. In this work, we propose DiffMOD, a framework that reformulates moving vehicle detection as a diffusion-based denoising process over sparse point hypotheses, where candidate locations are progressively refined from noise toward target centers. To adapt diffusion modeling to the high-sparsity remote sensing scenario, a density-guided initialization mechanism is introduced to guide hypothesis sampling toward informative regions using lightweight density estimation. A spatial-relation self-attention (SRSA) module is further designed to enhance interactions among sparse hypotheses and aggregate contextual spatio-temporal cues for recovering weak targets in cluttered scenes. In addition, a coverage-constrained assignment (CCA) strategy combined with a progressive training scheme is developed to stabilize optimization and gradually tighten supervision during training. Extensive experiments on the RsCar and SDM-Car benchmarks demonstrate the effectiveness of the proposed method. DiffMOD achieves state-of-the-art performance and improves the previous best method by 6.3% in $F1$-score and 13.8% in recall on the challenging SDM-Car dataset.**

***Index Terms*—Diffusion model, moving object detection (MOD), remote sensing, spatial-relation aggregation attention.**

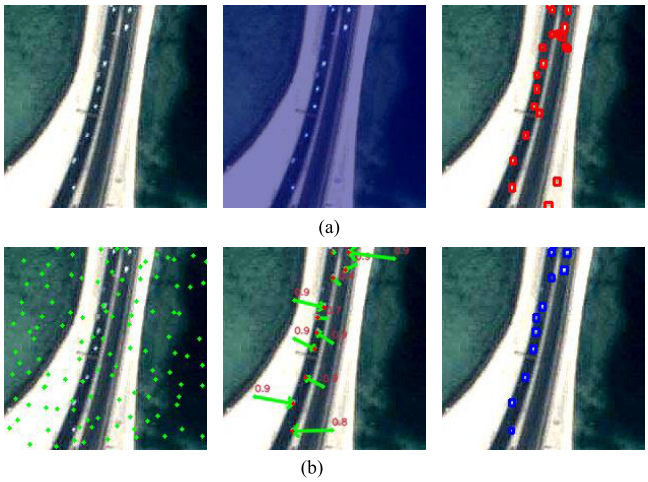


Fig. 1. Paradigm comparison between grid-aligned dense detection and the proposed diffusion-based point denoising framework. (Top) Conventional CNN-based detectors perform dense prediction over spatial grids, estimating object confidence at every location, which leads to overwhelming background responses and unstable localization for tiny vehicles. (Bottom) DiffMOD reformulates detection as a progressive diffusion-based denoising process over point hypotheses, where randomly initialized points are iteratively refined toward true target centers, enabling more robust recovery of small and weak moving objects in remote sensing videos. (a) Grid-aligned dense detection (DSFNet). (b) Diffusion-based point denoising (DiffMOD).

I. Introduction

REMOTE sensing video-based moving vehicle detection plays a critical role in a wide range of applications, including traffic monitoring, urban planning, and security surveillance [1], [2], [3], [4]. Compared with natural scene videos, remote sensing videos cover much larger geographic areas and exhibit complex background dynamics caused by parallax effects, illumination variations, and sensor noise [5], [6]. More importantly, moving vehicles in such videos usually occupy only a few pixels (e.g., less than 10×10), resulting in extremely low signal-to-noise ratios and severe foreground–background imbalance [7], [8]. These characteristics jointly make accurate detection of moving vehicles in remote sensing videos highly challenging.

Most existing deep-learning-based approaches address moving vehicle detection in remote sensing videos by extending convolution-based object detection paradigms to the temporal domain [9], [10], [11]. These methods [6], [12], [13], [14], [15], [16] typically rely on dense, grid-aligned prediction schemes, where object hypotheses are generated through pixel-wise confidence estimation or predefined bounding box priors.

While effective in many scenarios, such dense prediction faces fundamental limitations when confronting the extreme sparsity, diminutive scale, and complex backgrounds of satellite videos. As visually demonstrated in Fig. 1(a), estimating object confidence at uniformly distributed spatial grids under severe foreground–background imbalance inevitably produces overwhelming background responses and wastes computational resources. Furthermore, for tiny vehicles occupying only a few pixels, conventional grid-aligned anchors suffer from severe grid quantization errors. Even a slight spatial misalignment causes the extracted features to be completely

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dominated by background clutter. Coupled with static assignment strategies, it becomes exceedingly difficult for dense convolutional frameworks to progressively refine ambiguous, low-contrast targets.

To overcome these specific limitations, we propose DiffMOD, which reformulates moving vehicle detection as a progressive diffusion-based denoising process over sparse point hypotheses. This formulation is fundamentally more suitable for satellite video detection. Instead of relying on rigid, dense confidence maps, DiffMOD operates in a continuous coordinate space, explicitly bypassing grid quantization errors. As explicitly compared in Fig. 1(b), rather than being trapped by local background clutter in a fixed grid, DiffMOD begins with a set of sparse hypotheses and iteratively refines their coordinates toward true target centers. This diffusion-based denoising paradigm [17], [18], [19], [20] naturally offers a dynamic, coarse-to-fine search mechanism, enabling the model to continuously aggregate contextual cues and progressively recover weak targets from complex remote sensing backgrounds.

This paradigm shift brings two key advantages. First, point-based modeling leads to a more direct and well-conditioned optimization objective for tiny targets. By directly approximating target centers in a low-dimensional space, DiffMOD reduces the search space and avoids the sensitivity of dense prediction to small spatial deviations, which is particularly critical when objects occupy only a few pixels. This formulation aligns the learning objective more closely with the true localization goal and improves robustness under severe foreground–background imbalance. Second, sparse point hypotheses naturally enable structured feature interaction. Compared with largely independent dense predictions, sparse hypotheses can explicitly exchange information through spatial relationships, facilitating contextual aggregation and collaborative refinement. This property is particularly beneficial for recovering weak targets in cluttered remote sensing scenes, where individual observations are often ambiguous.

To address the above challenges, a novel framework named DiffMOD is proposed, which reformulates moving vehicle detection in remote sensing videos as a progressive diffusion-based denoising process over sparse point hypotheses. This formulation provides a new perspective for handling extreme foreground sparsity and tiny target ambiguity. Building upon this paradigm, a series of mechanisms tailored for diffusion-based sparse detection are further developed, including density-guided hypothesis initialization, spatial-relation-aware feature interaction, and coverage-constrained supervision, enabling robust recovery of weak targets in cluttered remote sensing scenes. Extensive experiments on the RsCar [6] and SDM-Car [21] benchmarks validate the effectiveness of the proposed approach, demonstrating clear advantages in detecting weak and small targets under severe foreground–background imbalance.

The main contributions of this work are summarized as follows.

1) We reformulate moving vehicle detection in satellite videos as a diffusion-based denoising process over sparse point hypotheses, providing an alternative to grid-aligned dense detectors and offering a principled solution to extreme foreground sparsity and tiny-target ambiguity.
2) A density-guided initialization mechanism is proposed that leverages lightweight density estimation to guide the sampling of initial hypotheses toward informative regions, improving the efficiency and stability of diffusion-based detection under highly sparse foreground conditions.
3) A spatial-relation self-attention (SRSA) module is developed to model interactions among sparse hypotheses and enhance contextual feature aggregation, enabling the recovery of weak targets in cluttered remote sensing scenes.
4) A coverage-constrained assignment (CCA) strategy together with a progressive training scheme is developed to stabilize optimization and gradually tighten supervision, leading to improved convergence and more robust localization.

II. Related Work

A. MOD in Remote Sensing

Compared with natural videos, satellite videos usually exhibit wider scenes, lower spatial resolution, extremely small object sizes, and complex background interference. These characteristics make moving object detection (MOD) particularly challenging when relying solely on appearance and texture cues. As a result, both model-based and learning-based methods have been developed to better exploit temporal information in satellite videos.

Early model-based methods typically formulate satellite videos as a combination of stable background, sparse moving targets, and noise. Representative approaches detect moving objects via frame differencing [5] or background modeling using mean or median filtering [22], [23]. However, such methods are highly sensitive to noise and background variations. To improve robustness, low-rank and sparse decomposition techniques [24], [25], [26], [27], [28], [29], [30] have been proposed to separate foreground targets from noise through global optimization. Despite their theoretical soundness, these methods suffer from high computational complexity and scale poorly to large-scale satellite video data.

With the advances of deep learning, learning-based MOD methods for satellite videos have emerged [6], [12], [13], [31], [32], [33], [34], [35], [36]. ClusterNet [31] adopts a two-stage strategy to first locate candidate regions in wide scenes and then detect small moving targets using appearance and motion cues. DSFNet [6] employs a two-stream architecture, combining a 2-D backbone for static context extraction and a lightweight 3-D backbone for dynamic motion modeling. GA-PANet [32] extracts motion information through historical frame differencing and fuses it with current frame features to obtain spatio-temporal representations. Recent studies [37], [38] highlight the importance of deeply exploring fine-grained spatial context and detection-driven feature enhancement for localizing extremely small or low-visibility objects. Furthermore, by fully utilizing temporal continuity and the intrinsic physical characteristics of targets, some frameworks [39], [40], [41] have successfully isolated moving objects from dynamic clutter using adaptive spatial–spectral representations and cross-modal prompts. Moreover, the continuous evolution of feature disentanglement modeling provides robust mechanisms to suppress complex background anomalies and

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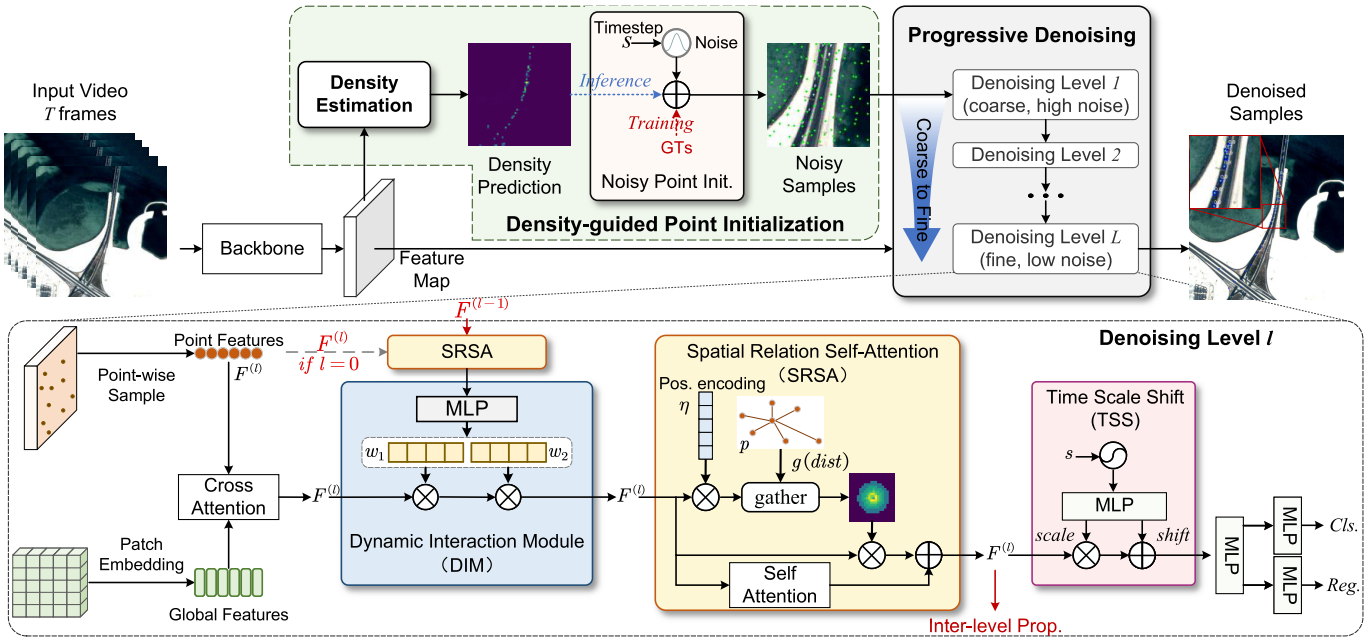


Fig. 2. Overview of the proposed DiffMOD framework. The backbone extracts shared spatio-temporal features from the input video. A density estimation branch predicts regional target density to guide noisy point initialization at inference, while ground-truth centers are used during training. The initialized noisy points are progressively refined through multiple denoising levels from coarse to fine, conditioned on backbone features, and the final denoised samples represent detected moving objects.

extract invariant target properties across broader observation domains [42], [43], [44], [45], [46]. Although these methods improve detection performance, they still rely on dense prediction over high-resolution feature maps, which leads to severe foreground–background imbalance, limited long-range interaction, and frequent missed detections for extremely small targets.

B. Sparse and Diffusion-Based Detectors

Recently, detection paradigms have increasingly shifted toward sparse modeling to avoid the heavy computational overhead of dense grids. Query-based frameworks, pioneered by DETR [47] and advanced by variants like Deformable DETR [48], DINO [49], and RT-DETR [50], formulate detection as a set prediction problem using statically learned queries. Alongside these, methods like sparse R-CNN [51] and Co-DETR [52] have iteratively refined sparse proposals and assignment strategies to enhance matching efficiency.

Concurrently, diffusion models have emerged as a powerful framework for generative modeling and representation learning [53]. By formulating data generation as an iterative denoising process from noisy samples, diffusion models provide a principled way to progressively refine complex data distributions [54], [55]. Beyond generative tasks, diffusion models have been extended to various discriminative applications, including multiobject tracking (MOT) [56], [57], and trajectory prediction [58], [59]. Advancing this trend into sparse detection, DiffusionDet [54] pioneers the application of diffusion models to object detection. In this framework, object queries are treated as continuous noisy instances, formulating detection as a progressive denoising process from noisy object boxes to ground-truth bounding boxes.

While these frameworks offer promising alternatives to dense prediction, they face fundamental limitations when confronting the extreme foreground–background imbalance and diminutive target sizes in satellite videos. In particular, applying box-based representation to extremely tiny targets introduces redundant degrees of freedom that exacerbate optimization instability. To address this, DiffMOD transitions to sparse point hypothesis modeling, which inherently better suits the diminutive scale of satellite targets and avoids box-related ambiguities. Moreover, static queries often struggle to efficiently capture the highly uneven target distributions in large-scale remote sensing scenes. In response, we introduce a dynamic density-guided initialization mechanism that adaptively allocates hypotheses to highly probable regions, significantly improving the recall of scattered targets. Finally, the strict one-to-one bipartite matching used in standard query-based decoders is notoriously unstable for heavily clustered, low-contrast tiny targets. To overcome this, DiffMOD proposes a coverage-constrained optimization strategy that guarantees robust spatial coverage and more stable convergence, effectively minimizing missed detections in complex backgrounds.

III. Method

A. Overview

Given a remote sensing video sequence, our goal is to detect moving vehicles in the current frame I_t. These targets are typically extremely small, densely clustered, and sparsely distributed across the vast scene. To address the fundamental limitations of dense detectors in such scenarios, we propose DiffMOD, a density-aware framework that formulates MOD as a progressive diffusion-based denoising process over a sparse set of point hypotheses. As illustrated in Fig. 2, the framework operates through the tight operational synergy of three core components: spatio-temporal feature extraction, density-guided point initialization, and progressive denoising.

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To provide a holistic understanding of DiffMOD, we detail how these modules cooperate dynamically during the training and inference phases.

1) Synergy During Training: The pipeline begins with a shared backbone [6] that extracts spatio-temporal feature maps from I_t, aggregating appearance and motion cues from adjacent frames. Based on these features, a density head predicts a region-level spatial probability distribution. This density-guided initialization effectively resolves the "cold start" sparsity problem by placing N initial point hypotheses in high-probability foreground regions, ensuring the computational budget is focused on informative areas. To simulate the generative diffusion process, ground-truth centers are perturbed with Gaussian noise conditioned on a time step s. These coarse hypotheses are then fed into the progressive denoising module, which consists of L cascaded decoder levels. Significantly, the CCA strategy acts as the dynamic supervisory bridge. At each internal level $l \in \{1, \ldots, L\}$, CCA evaluates the intermediately refined points, enforces strict spatial coverage constraints, and computes a coverage-aware miss loss. This level-by-level supervision tightly couples with the denoising module, forcing the network to progressively correct position errors and recover missed targets.

2) Synergy During Inference: During inference, the pipeline operates efficiently without the artificial noise injection or the CCA matching overhead. For sequential video frames, the density-guided initialization dynamically fuses the spatial density map of the current frame with temporal motion priors inherited from the previous frame. This robust initial hypothesis set undergoes a forward refinement pass through the L-level denoising module. Conditioned on the step s, the module progressively updates point coordinates to recover accurate target centers. The final refined outputs are then directly processed by nonmaximum suppression (NMS) to yield the ultimate moving vehicle trajectories.

B. Density-Guided Point Initialization

Standard random sampling is fundamentally inefficient for remote sensing imagery due to the overwhelming dominance of background regions. To efficiently cover sparse targets without relying on dense grids, we introduce a density-guided initialization strategy.

1) Region-Level Density Estimation: We first estimate a coarse object density map to serve as a sampling prior. The backbone features of frame I_t are partitioned into nonoverlapping patches via a patch embedding operation. A lightweight prediction head then processes these features to estimate the object count within each local region.

Considering that moving targets in remote sensing follow a zero-inflated distribution (i.e., most regions contain no targets), directly regressing counts can be numerically unstable. To address this issue, we decompose the prediction into two components: a soft presence probability map $D_{\text{mask}} \in [0, 1]^{H_p \times W_p}$ and a regression map $D_{\text{value}} \in \mathbb{R}^{H_p \times W_p}$ that predicts local object counts.

The final density map is computed via elementwise multiplication

$$D = D_{\text{mask}} \odot D_{\text{value}} \tag{1}$$

where $\odot$ denotes the Hadamard product.

This probabilistic gating formulation suppresses regression responses in background regions while preserving count predictions in areas with high object likelihood. The resulting density map $D \in \mathbb{R}^{H_p \times W_p}$ effectively highlights regions with high target probability, providing a robust reference for point allocation without requiring precise pixel-level supervision.

2) Density-Guided Noisy Point Generation: The initialization of the N point hypotheses employs distinct strategies for training and inference to balance optimization stability and detection robustness.

a) During training: Point initialization serves as the data source for the denoising process. To enable the model to recover targets from various noise levels, we construct the initial set by mixing target-anchored points and random background points. In particular, for each ground-truth center gt, we generate target-anchored points via Gaussian perturbation

$$p = gt + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}\left(0, \sigma_s^2 \mathbf{I}\right) \tag{2}$$

where σ_s corresponds to the noise scale of the current diffusion time step s. These are supplemented by uniformly sampled background points to fill the proposal budget N. To facilitate stable convergence, the mixing ratio and repetition frequency of positive samples are dynamically adjusted via a curriculum strategy (which will be detailed in Section III-D4).

b) During inference: Since ground-truth locations are unavailable, we utilize the predicted density map D to guide sampling. The hypotheses are primarily allocated to local regions proportional to their predicted density scores.

Unlike the training phase where the point count N must be strictly fixed to satisfy the tensor alignment constraints of parallel batching, our point-based denoising architecture inherently supports variable-sized inputs. During inference, we exploit this flexibility to accommodate a temporal propagation mechanism (detailed in Section III-E). Consequently, the budget N merely serves as a minimum capacity constraint; any remaining slots below this base budget are filled with globally uniform samples to maintain a robust hypothesis density across the scene.

C. Progressive Diffusion-Based Point Denoising

To recover accurate target centers from the initialized hypotheses, we design a progressive denoising module that refines points through L cascaded levels. Let $\{p_i^{(l-1)}, f_i^{(l-1)}\}_{i=1}^{N}$ denote the coordinates and feature vectors of the N point hypotheses input to the lth denoising level ($1 \le l \le L$). The backbone feature map is denoted as F. To obtain more comprehensive and stable scene context, we further derive patch features $F_g \in \mathbb{R}^{L_g \times d}$ from the same feature map F via a transformer-style patch embedding operation, which is identical to the feature construction used in the region-level density estimation stage, where $L_g = H_p \times W_p$.

As illustrated in Fig. 2, $\Phi_{block}^{(l)}$ updates the point features by sequentially sampling point features, integrating global context via cross-attention, interlevel history via dynamic interaction, and geometric relations via SRSA

$$f_i^{(l)}, p_i^{(l)} = \Phi_{block}^{(l)}\left(f_i^{(l-1)}, p_i^{(l-1)}, F, F_g, s\right). \tag{3}$$

First, local point features $f_i^{(l-1)}$ are bilinearly sampled from the feature map F at location $p_i^{(l-1)}$ for each point hypothesis.

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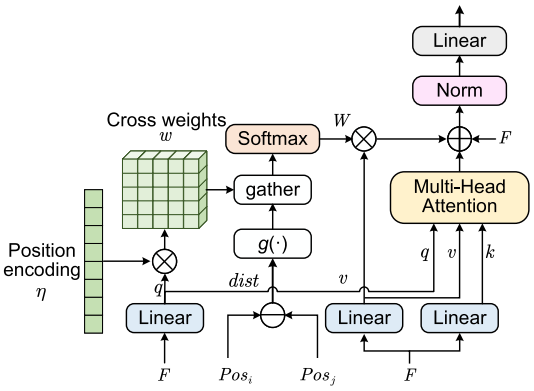
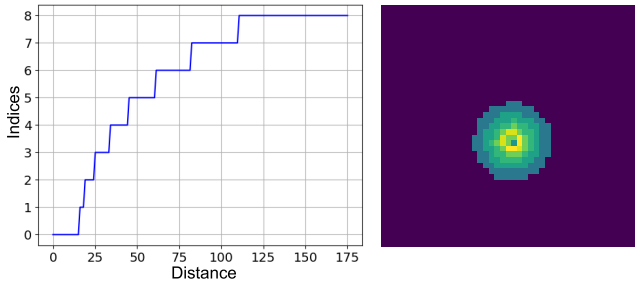


Fig. 3. Spatial-relation self-attention.



(a) Plot of the function $g(\cdot)$ (b) Attention Map

Fig. 4. Visualization of the distance quantization function $g(x)$ and the resulting SRSA. (a) Piecewise distance quantization function used to discretize relative spatial distances. (b) Example attention map illustrating distance-aware attention modulation, where nearby points receive stronger responses while distant points are progressively suppressed.

The region-level context features F_g, together with the point features, are then fed into the level-specific interaction block, where feature interaction is performed via the proposed SRSA, dynamic interaction module (DIM), and time-step scale-and-shift (TSS) modules to enhance spatio-temporal contextual aggregation. Finally, the updated features $f_i^{(l)}$ are processed by two parallel lightweight heads. A regression head $\mathcal{R}^{(l)}(\cdot)$ predicts coordinate offsets to refine the point positions

$$p_i^{(l)} = p_i^{(l-1)} + \mathcal{R}^{(l)}\left(f_i^{(l)}\right). \tag{4}$$

Simultaneously, a classification head $\mathcal{C}^{(l)}(\cdot)$ predicts the confidence scores $c_i^{(l)}$, indicating the likelihood of the hypothesis being a target

$$c_i^{(l)} = \mathcal{C}^{(l)}\left(f_i^{(l)}\right). \tag{5}$$

This hierarchical formulation allows DiffMOD to progressively refine both geometric localization and semantic classification of targets.

1) Spatial-Relation Self-Attention: Standard self-attention relies solely on semantic similarity, which can be highly unreliable for small targets that share appearance characteristics with background clutter. To mitigate this, we propose SRSA, which explicitly modulates attention weights using geometric distance to introduce a spatial prior that encourages locality, as illustrated in Fig. 3.

Given feature embeddings $F \in \mathbb{R}^{N\times d}$ and their spatial coordinates $Pos \in \mathbb{R}^{N\times 2}$ (where d is the feature dimension), we first compute the pairwise Euclidean distance matrix $dist \in \mathbb{R}^{N\times N}$. To prevent sensitivity to large spatial scale variations, the continuous distances are discretized into a stable structured representation using a piecewise quantization function $g(\cdot)$ (visualized in Fig. 4)

$$g(x) = \begin{cases} \left\lfloor \dfrac{x}{\alpha} \right\rfloor, & x \le \alpha \\ \min\left(\beta, \left\lfloor 1 + \dfrac{\ln(x/\alpha)}{\ln \gamma}(\beta-1) \right\rfloor\right), & x > \alpha \end{cases} \tag{6}$$

where α is the spatial stride controlling the local linear resolution, β specifies the total number of distance bins, and γ governs the logarithmic scaling. This maps the distances to $(\beta+1)$ discrete spatial levels, yielding the quantized index matrix $g(dist) \in \mathbb{R}^{N\times N}$.

Concurrently, the features F are projected to interact with a learnable relation position encoding $\eta \in \mathbb{R}^{(\beta+1)\times d}$ to compute cross-level similarity weights $w \in \mathbb{R}^{N\times(\beta+1)}$. Guided by the quantized indices $g(dist)$, the corresponding relation embeddings are gathered to construct the spatial relational attention matrix $W \in \mathbb{R}^{N\times N}$, achieving feature-dependent distance modulation.

Finally, W is normalized via a Softmax operation and applied to the value features. The resulting spatially aggregated representation is fused with the standard multihead self-attention output and the original input F through residual connections, followed by layer normalization and linear projection. Notably, for the initial denoising level, SRSA operates directly on the point features $F^{(0)}$ extracted at the current level, whereas subsequent levels inherit the refined features from the preceding stage (see Fig. 2).

2) Dynamic Interaction Module: To facilitate stable feature evolution, DIM manages interlevel information propagation. Since the feature distribution shifts as the diffusion process transitions from high-noise to low-noise states, direct stacking of layers may be suboptimal. As shown in Fig. 2, DIM takes the features from the previous level $f^{(l-1)}$ and fuses them with the current intermediate features. This fusion is governed by a gating mechanism conditioned on the diffusion step s, allowing the model to adaptively retain or update information based on the current noise intensity.

3) Time-Step Scale-and-Shift: TSS operation normalizes the point features $f^{(l)}$ using embeddings derived from s, ensuring the representation is aligned with the specific noise level before regression.

D. Assignment and Optimization Strategy

Training sparse detectors poses a unique challenge: the N point hypotheses may not spatially align with the sparse ground-truth targets, leading to instability. To address this, we employ a CCA strategy and a specialized optimization scheme.

1) Coverage-Constrained Assignment: Assigning sparse point hypotheses to ground-truth targets is particularly challenging in remote sensing scenarios, where objects are extremely small, sparsely distributed, and the number of hypotheses does not necessarily match the number of targets. Under such conditions, conventional assignment strategies often suffer from insufficient target coverage. To address this issue, a CCA strategy is introduced to enforce recall-oriented supervision.

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Algorithm 1 CCA at Level l

Input : Point set $P = \{p_i\}_{i=1}^{N}$, Ground-truth set $G = \{gt_j\}_{j=1}^{M}$, Radius r_l
Output: Assignment sets $\{\mathcal{A}_j\}_{j=1}^{M}$
Initialization:
// Compute Euclidean distance matrix (Cost Matrix)
Compute $E_{ij} \leftarrow \|p_i - gt_j\|_2$ for all i, j;
Initialize assignment sets $\mathcal{A}_j \leftarrow \emptyset$ for all $j = 1, \ldots, M$;
Set of unmatched targets $\mathcal{U} \leftarrow \{1, \ldots, M\}$;
Set of available points $\mathcal{V} \leftarrow \{1, \ldots, N\}$;
Stage 1: Iterative one-to-one coverage matching
while $\mathcal{U} \neq \emptyset$ **do**
  // Each unmatched target selects its nearest available point
  Candidate pairs $\mathcal{M}_{cand} \leftarrow \emptyset$;
  foreach $j \in \mathcal{U}$ **do**
    $i^* \leftarrow \arg\min_{i \in \mathcal{V}} E_{ij}$; // Find nearest point
    $\mathcal{M}_{cand} \leftarrow \mathcal{M}_{cand} \cup \{(i^*, j)\}$;
  // Conflict resolution: if a point is picked by multiple targets
  foreach *point* i *appearing in* $\mathcal{M}_{cand}$ **do**
    Find j^* among candidates targeting i with min distance;
    Add i to $\mathcal{A}_{j^*}$, remove j^* from $\mathcal{U}$, remove i from $\mathcal{V}$;
    // Block matched pair in E to prevent re-selection
    $E_{i,:} \leftarrow \infty, \quad E_{:,j^*} \leftarrow \infty$;
Stage 2: Neighborhood-based one-to-many expansion
foreach $i \in \mathcal{V}$ **do**
  // Check distance to ALL targets (using original coordinates)
  $j^* \leftarrow \arg\min_{j \in \{1,\ldots,M\}} \|p_i - gt_j\|_2$;
  if $\|p_i - gt_{j^*}\|_2 < r_l$ **then**
    // Assign point i to target j^* if within radius r_l
    $\mathcal{A}_{j^*} \leftarrow \mathcal{A}_{j^*} \cup \{i\}$;

The complete procedure at denoising level l is rigorously summarized in Algorithm 1. Let $P = \{p_i\}_{i=1}^{N}$ denote the set of point hypotheses, and $G = \{gt_j\}_{j=1}^{M}$ denote the ground-truth target centers. The algorithm computes a Euclidean distance cost matrix $E \in \mathbb{R}^{N \times M}$, where $E_{ij} = \|p_i - gt_j\|_2$, to sequentially determine the assignment sets $\{\mathcal{A}_j\}_{j=1}^{M}$ through two stages:

Stage 1 (Iterative One-to-One Coverage Matching): The primary goal is to ensure every ground-truth target obtains at least one match. In each iteration, every unmatched target in the working set $\mathcal{U}$ selects its nearest available point from set $\mathcal{V}$ based on E, forming a temporary set of candidate pairs $\mathcal{M}_{cand}$. To resolve conflicts where a point is selected by multiple targets, it is assigned exclusively to the closest target. The matched pairs are then blocked in matrix E (set to infinity) to prevent reselection. This greedy process repeats until all targets are covered ($\mathcal{U} = \emptyset$).

Stage 2 (Neighborhood-Based One-to-Many Expansion): After foundational coverage is established, any remaining unassigned point in $\mathcal{V}$ is evaluated for local expansion. It is appended to its absolute nearest target j^* if the original spatial distance $\|p_i - gt_{j^*}\|_2$ falls strictly within a level-dependent radius r_l.

Assignment is performed independently at each denoising level. To stabilize training, the regression loss is computed using assignments derived from the input points of the current level, thereby avoiding a circular dependency between coordinate prediction and matching. On the contrary, the classification and miss losses are computed using assignments derived from the output predictions, encouraging progressively improved confidence estimation and target coverage.

2) Coverage-Aware Miss Loss: Standard detection losses only penalize matched pairs, ignoring ground-truth targets that might be poorly covered in the geometric space. To explicitly mitigate such coverage failures, we introduce a coverage-aware miss loss.

Consistent with the definitions in Algorithm 1, let $G = \{gt_j\}_{j=1}^{M}$ be the set of ground-truth targets and $P = \{p_i\}_{i=1}^{N}$ be the set of point hypotheses at level l. For each target gt_j, its coverage is measured by the minimum Euclidean distance to the set of point hypotheses, directly derived from the distance matrix E

$$E_{\min}^{j} = \min_{i \in \{1,\ldots,N\}} E_{ij} = \min_{i \in \{1,\ldots,N\}} \left\| p_i - gt_j \right\|_2. \tag{7}$$

Using the level-dependent neighborhood radius r_l, the miss loss is defined as

$$L_{miss} = \frac{1}{M} \sum_{j=1}^{M} -\ln\left(\text{sigmoid}\left(\gamma \cdot \left(r_l - E_{\min}^{j}\right)\right) + \epsilon\right) \tag{8}$$

where M is the number of ground-truth targets, γ controls the sigmoid slope, and ϵ ensures numerical stability.

Intuitively, when at least one sampled point lies well within the target neighborhood ($E_{\min}^{j} < r_l$), the loss remains close to zero; as the minimum distance exceeds the neighborhood radius, the loss increases rapidly. This objective serves as a global force, pulling the point cloud distribution to ensure every target is covered within the radius r_l.

3) Overall Objective: The total training objective is a weighted sum of the classification loss (focal loss), regression loss (Smooth L1), and the proposed coverage-aware miss loss, aggregated across all L levels

$$L_{\text{total}} = \sum_{l=1}^{L} \left(\lambda_{\text{cls}} L_{\text{cls}}^{(l)} + \lambda_{\text{reg}}^{(l)} L_{\text{reg}}^{(l)} + \lambda_{\text{miss}} L_{\text{miss}}^{(l)}\right) \tag{9}$$

where λ_{cls} and λ_{miss} are shared across all levels, while the regression weight is level-dependent

$$\lambda_{\text{reg}}^{(l)} = \lambda_{\text{reg}} \cdot 2^{l-1}. \tag{10}$$

Meanwhile, the search radius at level l follows a progressively shrinking schedule:

$$r_l = r_0 \cdot 0.5^{l-1}. \tag{11}$$

This exponentially decaying design enforces a coarse-to-fine refinement mechanism: as the spatial search region contracts, the regression supervision is proportionally strengthened, thereby encouraging increasingly precise localization at deeper levels.

4) Progressive Training Strategy: To address the extreme sparsity and foreground–background imbalance, we implement a unified coarse-to-fine training strategy. This strategy progressively tightens supervision across two distinct phases of the training pipeline: point initialization and target assignment.

a) Curriculum sampling (for initialization): We propose a multistage curriculum to dynamically adjust the composition of initialized point hypotheses. Let ω_{pos} denote the positive sample ratio and $n_{\max}$ denote the maximum repetition limit for ground-truth targets. Given M ground-truth targets and a total budget of N points, the target number of positive samples is calculated as $N_{pos} = \lfloor N \cdot \omega_{pos} \rfloor$. The construction of the initial point set follows a conditional strategy.

1) *Sufficient Targets ($M > N_{pos}$):* If the number of ground-truth targets exceeds the budget, we utilize all M targets

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without repetition to maximize diversity. If the number of ground-truth targets exceeds the whole budget ($M > N$), we randomly sample N points from the M ground-truth targets without replacement.

2) *Sparse Targets ($M \leq N_{pos}$):* If targets are sparse, we replicate them to meet the demand N_{pos}, subject to the limit $n_{\max}$. The repetition count n_{rep} is determined by

$$n_{\text{rep}} = \min\left(\left\lfloor \frac{N_{pos}}{M} \right\rfloor, n_{\max}\right). \tag{12}$$

Each ground-truth target is repeated n_{rep} times. The remaining gap $N_{\text{remain}} = N_{pos} - n_{\text{rep}} \cdot M$ is filled by randomly sampling N_{remain} targets from the ground truths.

The selected centers are then perturbed by Gaussian noise $\mathcal{N}(0, \sigma_s^2\mathbf{I})$ to form the positive hypotheses. Finally, the remaining slots in the budget are filled with uniform background samples. As training progresses, both ω_{pos} and $n_{\max}$ are decayed stepwise. In the final stage, we set $n_{\max} = 1$ and reduce ω_{pos} to mimic the inference distribution, forcing the model to discriminate targets from complex backgrounds.

b) Spatial tolerance tightening (for assignment): Complementary to the sampling curriculum, we also tighten the spatial constraints in the label assignment phase (Algorithm 1). As the denoising process moves from coarse to fine levels, the matching radius r_l is decayed ($r_l = r_0 \cdot 0.5^{(l-1)}$). This ensures that early layers focus on rough region recall with looser tolerance, while deeper layers enforce precise subpixel localization during optimization.

Algorithm 2 Inference Pipeline of DiffMOD

Input : Video frames $\{\mathbf{I}_t\}_{t=1}^{T_{video}}$, Proposal budget N
Output: Detected target sets $\mathcal{D}_{final}$
for $t = 1$ **to** T_{video} **do**
  // 1. Feature Extraction
  Extract spatio-temporal features F_t from frame $\mathbf{I}_t$;
  // 2. Residual-based Initialization
  Predict density map D_t;
  Initialize point set $p_{init} \leftarrow \emptyset$;
  if $t > 1$ *and* $\mathcal{D}_{t-1} \neq \emptyset$ **then**
    // Inherit points and compute residual density
    Retrieve high-confidence centers p_{prev} from $\mathcal{D}_{t-1}$;
    Add p_{prev} to p_{init};
    // Suppress density at tracked locations
    $D_{residual} \leftarrow D_t - \text{Density}(p_{prev})$;
  else
    $D_{residual} \leftarrow D_t$;
  // Sample from residual density (Adaptive Size)
  Sample p_{new} from $D_{residual}$;
  Add p_{new} to p_{init};
  // Global random padding (Ensure Min Budget)
  if $|p_{init}| < N$ **then**
    $N_{pad} \leftarrow N - |p_{init}|$;
    Sample N_{pad} uniform random points p_{rand};
    Add p_{rand} to p_{init};
  // 3. Progressive Denoising (Single Pass)
  Pass p_{init} through L denoising levels to obtain refined coordinates and scores, yielding p_{out};
  // 4. Post-processing
  Filter points in p_{out} by confidence threshold τ_{cls};
  Apply NMS to generate final detections $\mathcal{D}_t$;
  Store $\mathcal{D}_t$ in $\mathcal{D}_{final}$;
return $\mathcal{D}_{final}$

E. Inference Procedure

Fundamentally, our point-based denoising architecture accepts point sets of arbitrary sizes. While a fixed point count N is enforced during training for batch processing, the inference phase operates on flexible point sets. We exploit this architectural flexibility to introduce a temporal propagation mechanism that leverages the strong spatio-temporal continuity of satellite videos.

The inference pipeline of DiffMOD is summarized in Algorithm 2. Given a test video sequence, we process it frame-by-frame. For each frame I_t, the backbone extracts spatio-temporal features, and the density head predicts a region-level density map D_t.

For the first frame $\mathbf{I}_1$ (or when no temporal prior exists), the network initializes the hypothesis set P_{init} directly based on D_1. For any subsequent frame $\mathbf{I}_t$ ($t > 1$), we explicitly incorporate temporal propagation. The high-confidence detections from the previous frame, denoted as P_{prev}, are retrieved from $\mathcal{D}_{t-1}$ and added to P_{init}. To prevent redundant sampling, we compute the residual density map $D_{residual} = D_t - \text{Density}(P_{prev})$, which effectively masks out the density mass around the inherited points.

The network then adaptively samples new hypotheses P_{new} from $D_{residual}$ and adds them to P_{init}. To guarantee a minimum hypothesis density and ensure robustness against sudden scene changes, we check if the combined size $|P_{init}|$ is less than the base budget N. If so, we calculate the deficit $N_{pad} = N - |P_{init}|$ and supplement the set with N_{pad} globally uniform random points, denoted as P_{rand}.

Once the flexibly sized hypothesis set P_{init} is prepared, it is fed into the progressive denoising module. To clarify the execution mechanism, we explicitly distinguish the internal architectural depth from the external inference cycles. The denoising module is fundamentally constructed as a cascade of L sequential decoder levels. During inference, we employ a "single refinement pass" strategy ($Iter = 1$), meaning that P_{init} propagates through this L-level cascade exactly once. Thanks to the highly accurate spatial priors provided by the density-guided temporal initialization, this single pass possesses ample capacity to achieve precise target localization. The predictions from the final Lth level, denoted as P_{out}, comprise the refined coordinates and their corresponding classification scores. Finally, NMS is applied to P_{out} using a threshold τ_{cls} to eliminate redundant predictions, yielding the ultimate moving vehicle detections $\mathcal{D}_t$ for the current frame.

IV. Experiments

A. Datasets and Metrics

To evaluate the effectiveness of DiffMOD, we conduct extensive experiments on two satellite video datasets: RsCar [6] and the SDM-Car [21] dataset.

1) RsCar Dataset: Collected by the Jilin-1 satellite [6], RsCar is a standard benchmark for satellite video object detection. The dataset focuses on moving vehicles as the primary targets. It consists of 79 video sequences, divided into 72 for training and 7 for testing. Each clip contains approximately 300 frames. The most significant characteristic of RsCar is the extreme smallness of the targets, with an average object size of only 8.5 $\times$ 6.7 pixels. During preprocessing, training images are cropped into 512 $\times$ 512 patches. For testing, the

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original resolution of 1024 × 1024 is maintained to preserve global context.

2) SDM-Car Dataset: The SDM-Car dataset [21] is acquired by the Luojia-3-01 satellite with a spatial resolution of 0.75 m. It comprises 99 video sequences (totaling 16 830 frames) with a resolution of 1024 × 1080 pixels. The dataset is partitioned into 64 sequences for training, 15 for validation, and 20 for testing. Compared to RsCar, SDM-Car presents significantly more complex challenges, including severe camera jitter and motion blur, denser and more complex object distributions, and diverse environmental interferences such as drastic lighting changes and cloud occlusions. Similar to RsCar, frames are cropped into nonoverlapping 512 × 512 patches during training, while inference is performed at the original resolution. Since the dataset is originally annotated for MOT, we filter out stationary objects during evaluation to align with the objective of MOD.

3) Evaluation Metrics: Following standard protocols [6], we evaluate the performance using precision, recall, and $F1$-score. Considering the extremely small size of targets (often <10 pixels), slight pixel-level deviations can cause drastic drops in intersection over union (IoU) despite accurate localization. Therefore, we set the IoU threshold to 0.1 for determining true positives, which is a common practice in satellite video object detection. The $F1$-score serves as the primary metric for comparison.

B. Implementation Details

1) Training Settings: We implement DiffMOD using the Detectron2 framework [60]. All models are trained on 4 NVIDIA Tesla V100 GPUs (32 GB memory). Following [6], we utilize the same backbone network for feature extraction, which takes a five-frame video clip as input to capture spatio-temporal information. The model is optimized using the AdamW optimizer with a total batch size of 32. The base learning rate is initialized at 5×10^{-4} for a total of 10 000 iterations. We employ a linear warm-up strategy for the first 1000 iterations. Subsequently, the learning rate is decayed by a factor of 0.7 at the 8000th iteration. The entire training process takes approximately 11 h.

2) Dynamic Training Strategy: To stabilize convergence under extreme sparsity, we adopt the dynamic curriculum strategy described in Section III-D4. The training process is divided into five stages (2000 iterations per stage). The hyperparameters for point initialization decay stepwise.

1) The maximum repetition limit $n_{\max}$ is set to $\{5, 5, 3, 3, 1\}$ for the five stages, respectively.
2) The positive sample ratio ω_{pos} is adjusted as $\{0.5, 0.25, 0.5, 0.25, 0.25\}$.

This schedule ensures high positive supervision in the early phases and mimics the sparse inference distribution in the final phase.

3) Inference Configurations: Unless otherwise specified, the number of initial noise points N is set to 500, and we use a single refinement pass ($Iter = 1$) for all experiments to balance accuracy and efficiency.

C. Main Results

To comprehensively evaluate the proposed DiffMOD, we benchmark it against two categories of approaches: 1) general small object detection (SOD) methods, including state-of-the-art single-stage detectors (RTMDet [61], YOLOv12 [62]) and the widely adopted slicing aided hyper inference (SAHI [63]) strategy and 2) MOD methods, encompassing traditional background modeling and the deep-learning-based baseline DSFNet [6]. Table I reports the quantitative comparison.

1) Performance on RsCar: On the RsCar dataset, DiffMOD achieves the best overall performance with an $F1$-score of 86.3%. Notably, the general SOD methods yield $F1$-scores only in the range of 53.1%–68.6%. This substantial performance gap stems from the fact that general SOD models process images frame-by-frame and lack temporal modeling, making them highly susceptible to false positives caused by static background objects. Furthermore, even when employing the state-of-the-art YOLOv12 as the base detector, the performance of the prevalent slicing strategy SAHI is noticeably degraded. Unlike low-altitude drone footage, satellite cameras capture scenes from extreme altitudes, meaning targets are universally small with minimal scale variation. Therefore, SAHI's multiscale zooming provides limited benefits. Worse still, its grid-based slice-and-merge operations inevitably fragment the global background topology, destroying the macroscopic spatial context that is critical for recognizing and filtering out target-like static distractors.

On the contrary, DiffMOD effectively exploits global spatiotemporal cues, outperforming DSFNet (83.7%) by 2.6%. This improvement highlights the capability of the proposed diffusion-based refinement to robustly filter out false positives induced by background clutter and noise. Simultaneously, DiffMOD maintains a highly competitive recall compared to the top-performing baseline, verifying that the sparse diffusion formulation does not compromise detection coverage in simpler scenes.

2) Performance on SDM-Car: The advantage of DiffMOD becomes even more evident on the highly challenging SDM-Car dataset. General SOD methods struggle severely, with YOLOv12 achieving an $F1$-score of 62.6%. Traditional background modeling methods fail to adapt to the camera motion. The deep learning baseline DSFNet achieves 76.6%. In stark contrast, DiffMOD maintains a robust $F1$-score of 82.9%. Notably, it improves recall by a large margin of 13.8% compared with DSFNet (82.6% versus 68.8%), demonstrating that the density-guided initialization and CCA strategy can effectively recover weak, blurred targets.

3) Inference Speed and Tradeoffs: In terms of computational efficiency, general SOD models exhibit extremely high frame rates (e.g., YOLOv12 at 30.7 FPS). However, their lack of temporal modeling makes them unsuitable for robust satellite video analysis. Furthermore, applying slicing strategies like SAHI to remote sensing images heavily degrades the speed to 6.4 FPS. DiffMOD operates at 2.0 FPS. Although slower than the lightweight DSFNet (5.0 FPS) due to iterative denoising, it remains significantly faster than traditional methods. Considering the remarkable 6.3% $F1$ gain over DSFNet on a challenging dataset, the computational cost is highly justified.

4) Visualization Analysis: To further qualitatively evaluate the effectiveness of DiffMOD, we present visualization comparisons with DSFNet shown in Fig. 5. The blue bounding

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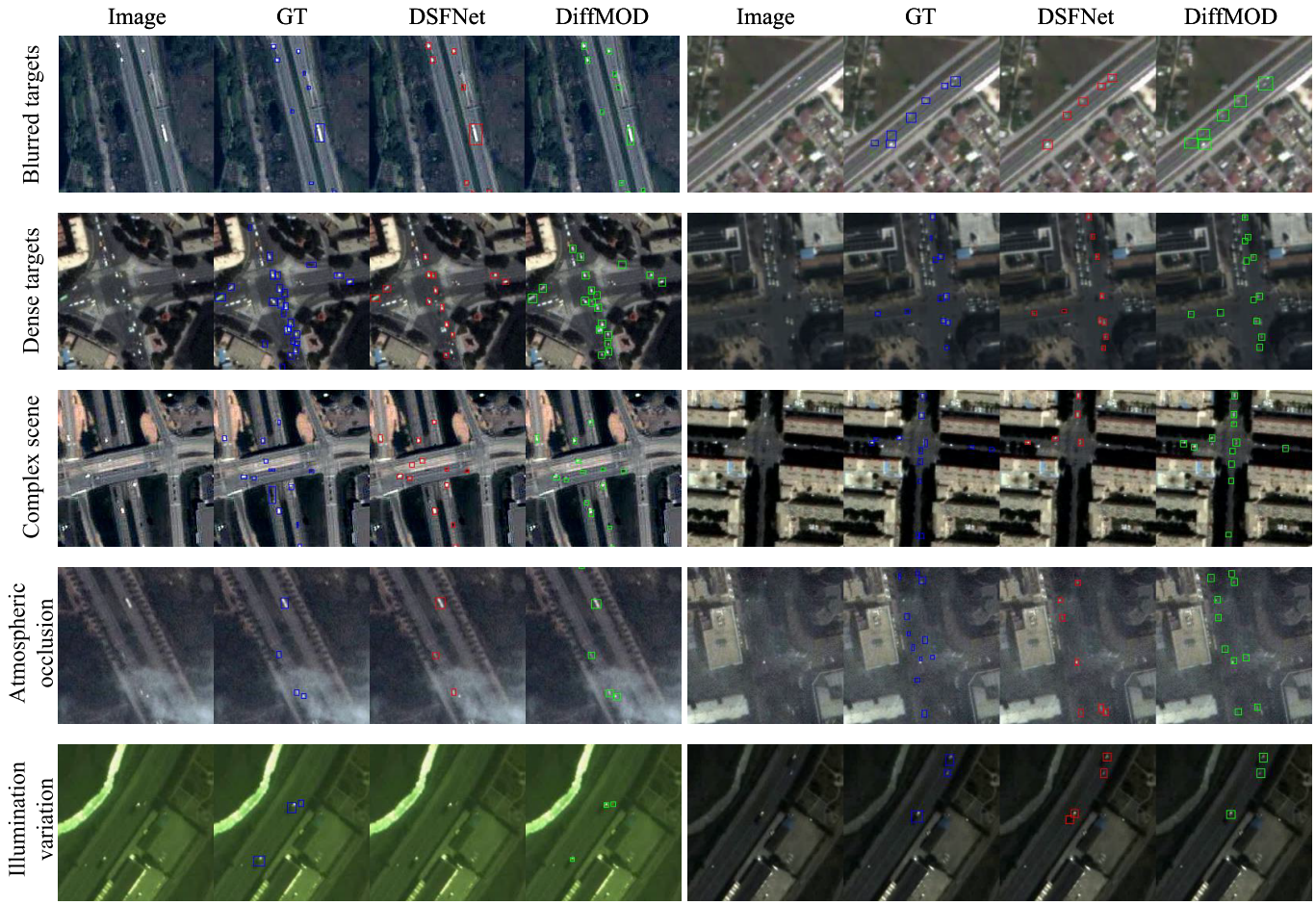


Fig. 5. Visualization comparisons of detection results on satellite videos. The blue boxes denote ground-truth annotations, while the red and green boxes denote the detected objects by DSFNet and DiffMOD. Small regions of interest are magnified for better visibility.

boxes denote ground-truth annotations, while the red and green ones indicate the detection results of DSFNet and DiffMOD, respectively. Representative scenarios with varying levels of complexity are included.

The first row illustrates cases where vehicle targets are severely blurred and difficult to distinguish from the background. DSFNet tends to miss low-contrast targets, whereas the proposed method successfully recovers them owing to its density-guided initialization and iterative refinement. The second row shows densely distributed vehicles in parking areas. In such crowded scenes, conventional detectors relying on dense, grid-aligned prediction schemes often struggle with overlapping responses. Our approach produces more complete detections, demonstrating improved coverage under high-density conditions. The third row presents challenging environments such as overpasses and intersections with partial occlusion from trees. Stronger robustness to structural clutter and occlusions can be observed, maintaining consistent detections where DSFNet produces fragmented or missing predictions. The fourth row corresponds to scenarios affected by atmospheric interference such as haze and cloud cover. Weak object responses are better preserved through contextual spatial interactions, leading to fewer missed detections. The final row illustrates scenes with significant illumination variations. Although a slight decrease in precision can be observed in some cases, substantially higher recall is achieved in complex environments, indicating a better balance between sensitivity and robustness.

Overall, these qualitative results demonstrate that DiffMOD is able to exploit contextual information more effectively, enhancing target recovery under challenging conditions while maintaining competitive precision.

D. Ablation Study

To further analyze the proposed framework, we conduct ablation studies on the challenging SDM-Car dataset. Unless otherwise specified, the default setting uses $N = 500$ points per patch and one denoising iteration.

1) Effectiveness of Architectural Modules: We evaluate the contribution of each architectural component by removing it from the full model. The results are summarized in Table II.

Removing SRSA and replacing it with standard multihead self-attention leads to a noticeable drop in $F1$ (82.9% $\rightarrow$ 81.8%). This result indicates that explicitly modeling spatial relations among sparse point hypotheses is beneficial for suppressing background interference and improving feature aggregation. Removing the TSS module results in the largest performance decrease ($F1 = 81.7\%$). This result indicates that scale awareness of the injected noise

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TABLE I

QUANTITATIVE COMPARISON ON RSCAR AND SDM-CAR DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Type	Method	RsCar			SDM-Car			FPS
		Recall	Prec	F1	Recall	Prec	F1	
SOD	RTMDet [61]	70.2	64.0	67.0	55.4	64.7	59.7	12.3
	YOLOv12 [62]	62.6	76.0	68.6	53.8	74.8	62.6	**30.7**
	SAHI [63]	67.4	43.8	53.1	50.4	73.8	59.9	6.4
MOD	VIBE [22]	68.8	34.7	45.7	65.7	29.3	40.5	0.8
	GMM [26]	53.6	55.2	54.4	46.2	45.5	45.9	-
	KNN [64]	62.1	58.7	60.4	58.3	41.2	48.3	0.2
	DSFNet [6]	85.3	83.2	83.7	68.8	**86.4**	76.6	5.0
	DiffMOD	**85.5**	**87.1**	**86.3**	**82.6**	83.3	**82.9**	2.0

TABLE II

ABLATION OF ARCHITECTURAL MODULES ON SDM-CAR

Variant	Recall	Prec	F1
w/o SRSA	82.2	81.5	81.8
w/o DIM	82.5	82.3	82.4
w/o Inter-level Prop.	81.9	82.0	81.9
w/o TSS	**82.8**	80.6	81.7
DiffMOD (Full)	82.6	**83.3**	**82.9**

TABLE III

IMPACT OF THE COVERAGE-AWARE MISS LOSS WEIGHT (λ_{miss}) ON RSCAR AND SDM-CAR DATASETS

Coverage-Aware Miss Loss Weight	RsCar			SDM-Car		
	Recall	Prec	F1	Recall	Prec	F1
0	84.6	**88.1**	**86.3**	82.3	82.8	82.6
2.5	**85.5**	87.1	**86.3**	82.3	82.9	82.6
5	**85.5**	86.5	86.0	**82.6**	**83.3**	**82.9**
7.5	84.9	85.8	85.3	82.4	81.3	81.9

is crucial in the denoising module. Disabling the DIM module reduces $F1\%$ to 82.4%, suggesting this module effectively facilitates feature interactions. Finally, removing the interlevel feature propagation decreases $F1\%$ to 81.9%. This confirms that progressive feature refinement across denoising levels is important for stable hypothesis evolution.

Overall, the full DiffMOD model achieves the best performance, demonstrating that the spatial, temporal, and progressive refinement components contribute complementarily to the final detection accuracy.

2) Impact of the Coverage-Aware Miss Loss Weight: We analyze the influence of the coverage-aware miss loss weight λ_{miss} on detection performance. The results are reported in Table III. Overall, the optimal weight depends on the target density of the dataset.

On the RsCar dataset, performance saturates at a relatively small weight of $\lambda_{miss} = 2.5$, where the model achieves the highest recall (85.5%) and $F1$-score (86.3%). Increasing the weight further leads to a gradual drop in precision without additional recall improvement, indicating that overly strong miss penalties may introduce false positives in sparse scenes.

On the SDM-Car dataset, the best performance is obtained at a larger weight of $\lambda_{miss} = 5$, achieving an $F1$-score of 82.9% with the highest precision (83.3%) and recall (82.6%). This result suggests that denser vehicle distributions require stronger coverage supervision to ensure sufficient target recall.

Based on these observations, we set $\lambda_{miss} = 2.5$ for RsCar and $\lambda_{miss} = 5$ for SDM-Car in the final model.

TABLE IV

ABLATION STUDY ON THE TWO-STAGE CCA

CCA	Recall	Prec	F1
Full Strategy	**82.6**	83.3	**82.9**
only Coverage Matching	72.8	74.0	73.4
only Neighborhood Expansion	80.6	**84.4**	82.4

TABLE V

COMPARISON OF LINEAR AND EXPONENTIAL SPATIAL TOLERANCE DECAY SCHEDULES

Schedule	Recall	Prec	F1
Exp	**82.6**	**83.3**	**82.9**
Linear	81.5	81.4	81.4

TABLE VI

EFFECT OF PROGRESSIVE TRAINING STRATEGY

ω_{pos}	n_{max}	Recall	Prec	F1
Progress		**82.6**	**83.3**	**82.9**
0.25	1	81.9	82.4	82.1
0.5	3	82.0	79.4	80.7

3) Effect of the Two-Stage CCA: Table IV evaluates the contribution of the proposed two-stage CCA.

Using only Stage 1 one-to-one matching (i.e., removing neighborhood expansion) leads to a substantial decrease in recall (72.8%) and $F1$-score (73.4%), indicating that one-to-one matching alone is insufficient to cover scattered targets in sparse point sets.

Conversely, using only neighborhood expansion (removing the Stage 1 matching) yields relatively high precision (84.4%) but reduced recall (80.6%). This result suggests that Stage 1 provides reliable anchor assignments, while Stage 2 improves coverage of nearby candidates.

Combining both stages achieves the best overall performance, demonstrating that the two components play complementary roles in balancing detection precision and recall.

4) Effect of Spatial Tolerance Decay Schedule: Table V illustrates the comparison of the linear and exponential spatial tolerance decay schedules used in progressive training. The exponential schedule achieves better overall performance ($F1$: 82.9% versus 81.4%), with consistent improvements in both recall and precision. This result indicates that progressively tightening the spatial tolerance at a faster rate in deeper denoising levels leads to more stable training and improved localization accuracy.

5) Effect of Progressive Training Strategy: Table VI lists the comparison of the proposed progressive training strategy with fixed training settings. Using a fixed easy setting ($n_{\max} = 3$, $\omega_{pos} = 0.5$) results in reduced Precision (79.4%) while using a fixed hard setting ($n_{\max} = 1$, $\omega_{pos} = 0.25$) leads to suboptimal convergence with a lower $F1$-score (82.1%). On the contrary, the progressive strategy achieves the best overall performance, indicating that gradually tightening the training difficulty provides a better balance between stable optimization and final detection accuracy.

6) Effectiveness of Density-Guided Initialization: Table VII evaluates the role of the proposed density-guided initialization. Using the density head only as an auxiliary training signal

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TABLE VII
EFFECT OF DENSITY-GUIDED INITIALIZATION

Train	Test	Recall	Prec	F1
		80.2	84.2	82.2
✓		80.3	**84.9**	82.6
✓	✓	**82.6**	83.3	**82.9**

TABLE VIII
SENSITIVITY ANALYSIS OF POINT BUDGET AND REFINEMENT ITERATIONS

Iter	num N	Recall	Prec	F1	FPS
1	300	82.3	**83.6**	**82.9**	**2.2**
1	500	82.6	83.3	**82.9**	2.0
2		82.7	83.1	**82.9**	1.6
3		82.7	83.1	**82.9**	1.4
1	700	**82.9**	82.9	**82.9**	1.8

improves $F1$ from 82.2% to 82.6%, indicating that density supervision enhances feature representation. Furthermore, utilizing the predicted density map for point sampling during inference increases $F1$% to 82.9%, suggesting that a better initial hypothesis distribution facilitates the subsequent denoising and refinement process.

E. Parameter Analysis

To comprehensively evaluate the operational efficiency and deployment feasibility of DiffMOD, we conduct a detailed parameter analysis. We first investigate the model's sensitivity to key inference hyperparameters—namely, the point budget (N) and the number of refinement iterations (*Iter*)—to determine the optimal speed–accuracy tradeoff. Subsequently, we provide a modular breakdown of the computational complexity to explore potential acceleration pathways for real-world remote sensing applications.

1) Robustness to N and Iter: As reported in Table VIII, detection performance remains remarkably stable across all tested settings, with the $F1$-score consistently reaching 82.9%. Increasing the point budget from $N = 300$ to $N = 700$ or expanding the external refinement iterations from $Iter = 1$ to $Iter = 3$ yields no additional $F1$ improvement. This observation strongly indicates that our density-guided initialization provides sufficiently accurate spatial priors, and that the internal L-level cascade within a single denoising pass possesses adequate refinement capacity, making external recycling iterations unnecessary.

2) Accuracy–Speed Tradeoff: Given the saturated detection performance observed above, the selection of hyperparameters is primarily driven by inference efficiency. Increasing the point budget from 300 to 700 slightly reduces the inference speed from 2.2 to 1.8 FPS. Meanwhile, expanding the number of refinement iterations imposes a more severe computational burden, dropping the speed to 1.4 FPS when $Iter = 3$. Considering this tradeoff, we adopt $N = 500$ and $Iter = 1$ as the default configuration, achieving an optimal balance between reliable target coverage and computational efficiency.

3) Computational Complexity and Acceleration Strategies: To explore acceleration pathways for practical deployment, we analyze the computational overhead of each module in DiffMOD. As detailed in Table IX, a clear structural bottleneck emerges: the feature extraction backbone consumes over 73% of the total inference time (0.39 s) and accounts for nearly 80% of the parameters (16.56M). In stark contrast, the core diffusion denoising module, operating strictly on sparse point hypotheses, requires merely 0.75M parameters and 0.08 s, demonstrating exceptional computational efficiency.

TABLE IX
MODULAR TIME CONSUMPTION AND PARAMETER BREAKDOWN OF DIFFMOD

Module	Time (s)	Parameters (M)
Feature Extraction (Backbone)	0.39	16.56
Point Initialization	0.02	3.32
Denoising Module	0.08	0.75
Post-processing	0.04	0.00
DiffMOD (Total)	**0.53**	**20.64**

Given this extreme imbalance, a highly effective acceleration pathway is computational decoupling. In particular, the heavy feature extraction can be optimized into an asynchronous, multiframe parallel process to fully exploit hardware concurrency and hide latency. Meanwhile, the subsequent lightweight point initialization and progressive denoising modules maintain serial execution. This decoupled architecture effectively circumvents the speed bottleneck of the dense backbone while strictly preserving the diffusion model's fine-grained capacity for temporal consistency and detection accuracy.

V. CONCLUSION

This article presented DiffMOD, a diffusion-based framework for moving vehicle detection in satellite videos. By reformulating detection as a progressive denoising process over sparse point hypotheses, the proposed approach provides an alternative to conventional grid-aligned dense detectors for handling extremely small targets and severe foreground–background imbalance. To adapt diffusion modeling to remote sensing scenarios, we introduced density-guided initialization for efficient hypothesis generation, a spatial-relation self-attention module for contextual feature aggregation, and a CCA strategy with progressive training to stabilize optimization. Extensive experiments on the RsCar and SDM-Car benchmarks demonstrated consistent improvements over existing methods, with particularly notable gains in recall under challenging conditions involving motion blur, camera jitter, and complex backgrounds.

Future work will focus on incorporating richer temporal modeling beyond backbone-level feature extraction, enabling more explicit motion reasoning and long-range temporal interaction for further improving detection robustness in satellite videos.

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